

Reminders and announcements

- ▶ During lectures you can submit questions to:
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Revision

- ▶ A **proposition** is something that has truth value of either **true (T)** or **false (F)**.
- ▶ The **logical operators** are $\neg$ (**NOT**), $\wedge$ (**AND**), $\vee$ (**OR**), $\Rightarrow$ (**IF THEN**), $\Leftrightarrow$ (**IFF**).
- ▶ In the implication $A \Rightarrow B$, A is called the **antecedent** and B is called the **consequent**.
- ▶ $A \Rightarrow B$ is false only if the antecedent is true and the consequent is false.

Equivalence

Definition 2.3. Let A and B be statements. If $A \Leftrightarrow B$ is always true, then A and B are **equivalent**.

Proposition 2.5. The following statements are equivalent.

- ▶ $A \vee \mathbf{T}$ and $\mathbf{T}$
- ▶ $A \wedge \mathbf{T}$ and A
- ▶ $A \vee \mathbf{F}$ and A
- ▶ $A \wedge \mathbf{F}$ and $\mathbf{F}$

Equivalence

Proposition 2.6. The following statements are equivalent.

Double negation:	A and $\neg(\neg A)$
Commutative laws:	$A \wedge B$ and $B \wedge A$ $A \vee B$ and $B \vee A$
Associative laws:	$A \wedge (B \wedge C)$ and $(A \wedge B) \wedge C$ $A \vee (B \vee C)$ and $(A \vee B) \vee C$
Distributive laws:	$A \vee (B \wedge C)$ and $(A \vee B) \wedge (A \vee C)$ $A \wedge (B \vee C)$ and $(A \wedge B) \vee (A \wedge C)$
Idempotent laws:	$A \wedge A$ and A $A \vee A$ and A
De Morgan's laws:	$\neg(A \wedge B)$ and $(\neg A) \vee (\neg B)$ $\neg(A \vee B)$ and $(\neg A) \wedge (\neg B)$

Equivalence

Proposition 2.7. The following statements are equivalent.

- ▶ $A \Rightarrow B$ and $(\neg A) \vee B$
- ▶ $\neg(A \Rightarrow B)$ and $A \wedge (\neg B)$

Predicate logic

Our next language uses **predicates** and **variables**.

We use symbols like $x, y, z, x_1, x_2, x_3, \dots$ as variables. These represents elements of the **domain**, which is the set of objects we are talking about.

For example, the domain may be the set of positive integers, non-negative integers, integers, rational numbers, or real numbers.

A predicate is represented by a symbol like P, Q , or E . They act like functions which take an element of the domain as input and produce a truth value as output.

Predicate logic

Assume our domain is the set of positive integers.

Let $P(x)$ be the predicate which will return the value **T** if x is prime, and **F** if x is non-prime.

(Recall that a positive integer is **prime** if it is divided by *exactly two* positive integers: 1, and itself.)

$P(3)$ is true $P(4)$ is false $P(13)$ is true $P(1)$ is false

Predicate logic

Assume our domain is the set of positive integers.

Let $D(x, y)$ be the predicate which will return the value **T** if x divides y and **F** otherwise.

$D(5, 10)$ is true $D(5, 11)$ is false

Quantifiers

The symbol $\exists$ is the **existential quantifier**.

We read

$\exists x$ blah blah blah

as “there exists x , an element of the domain, such that **blah blah blah** is true”.

Quantifiers

The symbol $\forall$ is the **universal quantifier**.

We read

$\forall x$ blah blah blah

as “for every element x in the domain, blah blah blah is true”.

Quantifiers

Say that the domain is the set of positive integers.

$P(x)$ means x is prime

$D(x, y)$ means x divides y

“For every positive integer x , there exists a prime number y such that y divides x ”.

Quantifiers

Say that the domain is the set of positive integers.

$P(x)$ means x is prime

$D(x, y)$ means x divides y

$$\forall x \forall y ((P(x) \wedge P(y) \wedge D(x, y)) \Rightarrow D(y, x))$$

Negating quantifiers

The negation of $\forall x$ **blah** is **not** the same as $\forall x \neg$ **blah**.

Negating quantifiers

Proposition 2.8. $\neg(\forall x \text{ blah})$ is equivalent to $\exists x \neg \text{blah}$.

Proposition 2.9. $\neg(\exists x \text{ blah})$ is equivalent to $\forall x \neg \text{blah}$.

Negating quantifiers

Tänk på att alla tåg inte
stannar på alla stationer.

Se opmærksom på, at alle tog
ikke standser på alle stationer.

Please note that all trains
do not stop at all stations.

Negating quantifiers

Show that $\neg \exists x (P(x) \Rightarrow \forall y Q(y))$ is equivalent to $\forall x (P(x) \wedge \exists y \neg Q(y))$.

Logical arguments

An **argument** consists of a sequence of statements, called the **premises**, and another statement, called the **conclusion**.

If $A_1, A_2, \dots, A_n$ are the premises and C is the conclusion, we represent the argument with the notation

$$A_1, A_2, \dots, A_n \therefore C.$$

Valid arguments

The point of a valid argument is that if we accept the premises as being true, then we must accept that the conclusion is true.

Definition 3.2. If the conclusion is true whenever all the premises are true, then the argument is **valid**.

Say that $A_1, A_2, \dots, A_n \therefore C$ is an argument. The argument is valid if and only if

$$(A_1 \wedge A_2 \wedge \dots \wedge A_n) \Rightarrow C$$

is always true (a **tautology**).

Valid arguments

An argument being valid doesn't necessarily mean that the premises are true!

Valid arguments

Theoretically, we can test that an argument is valid by writing the truth tables.

We check the rows where *all* the premises are true. In order for the argument to be valid, the conclusion must be true in *all* these rows.

Consider the argument $P_1 \Rightarrow (\neg P_2), P_2 \therefore \neg P_1$.

P_1	P_2	$P_1 \Rightarrow (\neg P_2)$	P_2	$\neg P_1$
T	T	F	T	F
T	F	T	F	F
F	T	T	T	T
F	F	T	F	T

Valid arguments

We check the rows where *all* the premises are true. In order for the argument to be valid, the conclusion must be true in *all* these rows.

Consider the argument $(P_1 \wedge P_2) \Rightarrow P_3, \neg P_2 \therefore (\neg P_1) \Rightarrow P_3$.

P_1	P_2	P_3	$(P_1 \wedge P_2) \Rightarrow P_3$	$\neg P_2$	$\neg P_1$
T	T	T	T	F	F
T	T	F	F	F	F
T	F	T	T	T	F
T	F	F	T	T	F
F	T	T	T	F	T
F	T	F	T	F	T
F	F	T	T	T	T
F	F	F	T	T	T

Valid arguments

Consider the argument $P_1, \neg P_1 \therefore P_2$.

P_1	P_2	P_1	$\neg P_1$	P_2
T	T	T	F	T
T	F	T	F	F
F	T	F	T	T
F	F	F	T	F

There are no rows where all the premises are all true. This makes the argument valid!

Rules of inference

Checking truth tables is impossible if there are a lot of atomic propositions, or if we are using predicate logic.

Instead, we use **rules of inference** to verify that an argument is valid.

We start by listing the premises of the argument.

We use the rules to produce logical statements from the statements that already appear in the sequence.

The rules are designed so that if we assume the premises are true, then every statement in the sequence is true.

If we eventually produce the conclusion of the argument, then we know that the truth of the premises forces the truth of the conclusion, so the argument is valid.

Conjunction elimination

Let A and B be any statements in propositional logic.

If $A \wedge B$ appears in the sequence, the

conjunction elimination rule lets us place A on a later line.

We can also place B on a later line.

If $A \wedge B$ appears on line j of the sequence, we make the note

$\wedge\text{Elim } j$

next to any line where we apply the conjunction elimination rule to deduce either A or B .

Conjunction introduction

Let A and B be any statements in propositional logic.

If A and B both appear in the sequence, the **conjunction introduction** rule lets us place $A \wedge B$ on a later line.

If $A \wedge B$ appears on line j of the sequence and B appears on line k , we make the note

$$\wedge\text{Intro } j, k$$

next to any line where we apply the conjunction introduction rule to deduce $A \wedge B$.

Fitch diagrams

Use the rules of inference to show that
 $P_1 \wedge P_2, P_3 \therefore P_1 \wedge P_3$ is a valid argument.

1.	_____
2.	_____
3.	_____
4.	_____

Implication elimination

Let A and B be any statements in propositional logic.

If $A \Rightarrow B$ and A appear in the sequence, the **implication elimination** rule lets us place B on a later line.

If $A \Rightarrow B$ appears on line j of the sequence and A appears on line k , we make the note

$\Rightarrow$ Elim j, k

next to any line where we apply the implication elimination rule to deduce B .

This rule is sometimes called **modus ponens**.

Example

Use the rules of inference to show that

$P_1 \Rightarrow (P_2 \wedge P_3)$, $P_1 \Rightarrow (P_3 \wedge P_4)$, $P_1 \therefore P_2 \wedge P_4$ is a valid argument.

1.

2.

3.

4.

5.

6.

7.

8.
